

Challenge (Habanero)

- Q1.** A hiker stands at marker m . If the distance between m and marker -14 on the trail is exactly 26 units, what is the sum of all possible values of m ?
- (A) -28
(B) -14
(C) 0
(D) 12
(E) -52
- Q2.** An expedition vehicle's fuel efficiency f deviates from its optimal 8.5 km/L by no more than 1.3 km/L. Which inequality represents the range of f ?
- (A) $|f + 8.5| \leq 1.3$
(B) $|f - 1.3| \leq 8.5$
(C) $|f - 8.5| \leq 1.3$
(D) $|8.5 - 1.3| \geq f$
(E) $|f + 1.3| \leq 8.5$
- Q3.** During a drought, the water level w of a lake compared to the sea level satisfies $||w| - 12| = 5$. If the lake is known to be below sea level ($w < 0$), what is the product of all possible values of w ?
- (A) -60
(B) 35
(C) 119
(D) 84
(E) -17
- Q4.** A rare bird species is found at an altitude h such that $100 < |h - 500| < 200$. Which of the following is a possible value for h ?
- (A) 350
(B) 450
(C) 650
(D) 750
- (E) 550
- Q5.** Evaluate the expression $|a - b| - |b + c|$ where a is the depth of a canyon (-150m), b is the height of a hill (40m), and c is the elevation of a base camp (110m).
- (A) 40
(B) 150
(C) 340
(D) 0
(E) 110
- Q6.** If the distance between a mountain goat at position x and a salt lick at position 12 is less than 5 , which of the following must be true?
- (A) $x < 7$
(B) $x > 17$
(C) $7 < x < 17$
(D) $-17 < x < -7$
(E) $x > 12$
- Q7.** Let x be the temperature in Celsius. If $|x + 10| = |x - 20|$, what is the physical interpretation of x ?
- (A) The temperature is equidistant from -10 and 20 .
(B) The temperature is 30 degrees.
(C) The temperature is always positive.
(D) The distance from zero is 10 .
(E) The temperature is exactly 15 degrees.
- Q8.** A salmon swims at a depth d (where $d < -5$). Simplify the expression $|d + 2| + |4 - d| - |d|$.
- (A) $6 - d$
(B) $d + 6$
(C) $2 - d$
(D) $2 + d$
(E) 6

Open Ended Questions (FRQ)

- Q9.** A research submarine is at a depth of -450 meters. It detects a thermal vent at depth v . If the distance between the submarine and the vent is $|-450 - v| = 125$ meters, determine all possible coordinate values for v and state which one is closer to the surface.
- Q10.** In a forest, the population of a specific tree species P is modeled by the equation $|2P - 800| = 150$. Solve for P and explain why the absolute value definition ensures there are two distinct population scenarios in this model.

Chef's Tips

- Remind students that $|a - b|$ always represents the distance between a and b on the number line.
- Encourage drawing a vertical number line for altitude and depth problems.
- Check for extraneous solutions when solving nested absolute value equations.